

3(a)	$E = \text{stress} / \text{strain} \text{ or } (F / A) / (e / l)$	C1
	$= [\text{gradient} \times 3.5] / [\pi \times (0.19 \times 10^{-3})^2]$ e.g. $E = \{[(40 - 5) / ([11.6 - 3.2] \times 10^{-3})] \times 3.5\} / [\pi \times (0.19 \times 10^{-3})^2]$ or $[4170 \times 3.5] / [\pi \times (0.19 \times 10^{-3})^2]$	C1
	$E (= 1.3 \times 10^{11}) = 0.13 \text{ TPa (allow answers in range 0.120–0.136 TPa)}$	A1
3(b)	a larger <u>range</u> of F required or <u>range</u> greater than 35 N	B1